

Chapter 109 — Neuromuscular Dysfunction of the Lower Urinary Tract

OBJECTIVES

- **Neuropathic lower urinary tract (LUT) dysfunction is largely predictable** from the site and nature of the lesion plus the physiology of storage and emptying — usually, but not always.
- **Secondary factors** (below) explain why storage and voiding abnormalities **persist after the precipitating cause has resolved**.
- Each disorder should be described by five **urodynamic parameters** (International Continence Society terms):
 - **Sensation** — normal, absent, impaired.
 - **Detrusor activity** — normal, overactive, areflexic, impaired contractility.
 - **Compliance** — normal, decreased, increased.
 - **Smooth sphincter** — synergic, dyssynergic.
 - **Striated sphincter** — synergic, dyssynergic, bradykinetic, impaired voluntary control, fixed tone.

TABLE 109.1 MOST COMMON PATTERNS OF VOIDING DYSFUNCTION SEEN WITH VARIOUS TYPES OF NEUROLOGIC DISEASE OR INJURY

Level	Disorder	Detrusor	Compliance	Smooth sphincter	Striated sphincter	Note
Above brainstem	CVA, brain tumour	Ov	N	S	S; $\pm$ VC (CVA)	Sensation of LUT events may be reduced
	Cerebral palsy	Ov	N	S	S; D in 25% of those with DO; $\pm$ VC	
	Parkinson disease	Ov / I	N	S	S; bradykinesia	
	Multiple system atrophy	Ov / I	N / $\downarrow$	Op	S	Striated sphincter may show denervation
Spinal cord	Multiple sclerosis	Ov	N	S	S; D in 30–65%	D figure = % of those with detrusor activity
	Suprasacral SCI	Ov	N	S	D	Smooth sphincter D if lesion above T7
	Sacral SCI	A	N; $\downarrow$ may develop	CNR; Op may develop	F	
	Autonomic hyperreflexia	Ov	N	D	D	
	Myelodysplasia	A	N / $\downarrow$	Op; $\downarrow$ may develop	F	Varies widely; striated denervation common
	Tabes, pernicious anaemia	I / A	N / $\uparrow$	S	S	Sensory loss primary ; detrusor decompensates from overdistension
Distal to cord	Disc disease	A	N	CNR	S	Striated denervation, fixed tone possible

Level	Disorder	Detrusor	Compliance	Smooth sphincter	Striated sphincter	Note
	Radical pelvic surgery	I / A	↓ / N	Op	F	
	Diabetes	I / A / Ov	N / ↑	S	S	Sensory and motor neuropathy

Key: A, areflexia; CNR, competent non-relaxing; D, dyssynergic; F, fixed tone; I, impaired; N, normal; Op, open (incompetent at rest); Ov, overactive; S, synergic; ±VC, voluntary control may be impaired; ↓ decreased; ↑ increased; CVA, cerebrovascular accident; SCI, spinal cord injury; DO, detrusor overactivity.

- **What the table is for:** predicting the pattern from the lesion level. **Suprapontine** → **DO with synergic sphincters**; **suprasacral cord** → **DO with striated dyssynergia**; **sacral/infrasacral** → **areflexia, often with low compliance, open bladder neck and fixed striated tone**.
- **Pattern across rows:** **dyssynergia appears only once the lesion lies between the pons and the sacral cord**; an **open smooth sphincter** marks autonomic (MSA, radical pelvic surgery, myelodysplasia) or sacral damage.
- **Trap:** **Parkinson disease and MSA share a row pattern for the detrusor but not for the outlet** — the open bladder neck and denervated striated sphincter belong to MSA.

PATTERNS OF NEUROPATHIC VOIDING DYSFUNCTION

- Discrete lesions affect filling/storage and emptying/voiding in a **consistent manner**, determined by: (1) **the area affected**; (2) **its function and contents**; (3) **whether the process is destructive, inflammatory or irritative**.
- **Acute dysfunction may differ from chronic**.

Neuroplasticity

- **Plasticity** — inherent capacity for structural and functional modification (morphologic, neurochemical, electrical, organisational); **reversible up to a point, then fixed** — explains **persistent symptoms after the initial stimulus**.
- **Most obvious examples:** reorganisation of the micturition reflex **after complete cord transection above S2**, and peripheral reorganisation after **damage to the parasympathetic innervation**.
- **Bladder outlet obstruction (BOO) as a plastic process:**
 - Stretch from overdistension or **ischaemia** → molecular events → **↑ contractile protein synthesis and hypertrophy**; excitation-contraction proteins altered; **emptying initially preserved**.
 - Extracellular matrix: **↑ type III : type I collagen ratio**, type III relocates within bundles → **↓ compliance**; ischaemia (obstruction or atherosclerosis) promotes fibrosis.
 - **Partial denervation** of intrinsic innervation (pressure + ischaemia).
 - **Storage failure** = matrix change (↓ compliance) + **phasic DO** — myogenic (partial denervation) or neurogenic (**afferent plasticity mediated by nerve growth factor [NGF]**; abolished experimentally by anti-NGF immunisation).
 - **Emptying failure** = reversal of compensation or breakdown of calcium handling (uptake, storage, release).
 - Fetal-sheep obstruction model: shift in muscarinic, purinergic and nitrergic mechanisms, hypocontractility (possible **atropine resistance**), **loss of the urothelial nitric-oxide-mediated negative inotropic effect**, reduced compliance.

- **Persistence after relief of obstruction:** ice-water test (**C-fibre-mediated**) more often positive in BOO → **primitive reflex circuitry**; persistent **afferent neuroplasticity explains symptomatic failures after urodynamically successful de-obstruction**. Rat data: DO that develops in BOO may be adaptive and normalises (with gene expression) after de-obstruction.
- **Retrograde neurotrophic signalling** alters spinal cord protein expression after SCI and irritant cystitis — may explain persistence after the stimulus is removed (e.g. **interstitial cystitis**).

Concept — *the lesion starts the dysfunction but plasticity maintains it: **afferent sensitisation and matrix remodelling outlive their cause**, so removing the trigger (obstruction, irritant) does not guarantee resolution.*

DISEASES AT OR ABOVE THE BRAINSTEM

Cerebrovascular Disease

Cerebrovascular Accident (Stroke)

- **Epidemiology:** a leading cause of death and disability; **~795,000 per year in the USA, 15 million worldwide**; prevalence **~60/1000 at ≥65 years, 95/1000 at ≥75 years**; ~1 in 200 will have a CVA; **fifth leading cause of US death**, but **~75% survive**, often with urgency, frequency and incontinence.
- **Causes:** thrombosis, occlusion, haemorrhage → infarction, usually **around the internal capsule**. Elderly patients bring **pre-existing LUT pathology**; cognitive impairment, dysphasia, immobility and low motivation complicate management.
- **Urinary incontinence (UI): 32–79% on admission → 25–28% at discharge → 12–19% months later.**
 - **UI persisting 7 days after stroke predicts poor survival and dependence**; UI on admission independently predicts 3-month death (confounded by infection and malnutrition).
 - Associated with **age ≥75, dysphagia, visual field defect, motor weakness**.
 - Lenticulocapsular strokes → **emotional incontinence**.
 - UI + faecal incontinence is the commonest pattern entering rehabilitation (~1/3).
 - **Commonest UI type = impaired-awareness incontinence** (insensate/delayed awareness) > pure urgency UI (UUI); **impaired awareness carries the worse prognosis**.
- **Acute phase: retention from detrusor areflexia ("cerebral shock")**, mechanism unclear → fixed deficit emerges over weeks–months.
- **Chronic pattern: phasic DO with intact sensation** (urgency, frequency) and **synergic smooth and striated sphincters**; if the patient can voluntarily contract the striated sphincter → urgency/frequency only; if not → UUI.
 - Lesions **confined to basal ganglia or thalamus → normal voluntary sphincter control**; lesions of **cortex and/or internal capsule → cannot forcefully contract the sphincter**. "**Uninhibited sphincter relaxation**" is a **misnomer** — the defect is in corticospinal voluntary control.
 - Hemispheric side (dominant, non-dominant, bilateral) does **not** change urodynamic findings.
 - Two UI mechanisms: **impaired striated sphincter control and lack of appreciation of filling/impending contraction**.
- **True detrusor–striated sphincter dyssynergia (DSD) does not occur**; reported "dyssynergia" is **pseudodyssynergia** — electromyographic (EMG) flare from voluntary attempts to inhibit an involuntary contraction. **Guarding reflex intact**.
- **Persistent areflexia is rare** (estimates up to 20%); post-stroke retention predicts poor function and recovery.

- **Men with pre-stroke LUTS:** poor flow/high post-void residual (PVR) usually = prostatic obstruction, but **full urodynamics before outlet surgery** to exclude **DO with impaired contractility**.
- **Drug trap:** antimuscarinics crossing the blood–brain barrier (**M1 blockade**) worsen confusion.
- **Symptom burden 1 month post-stroke:** ≥ 1 LUTS in most; **nocturia commonest**, then urgency, frequency; **urgency most bothersome**; bother tracks lower-limb paresis.
- **Mechanism of DO:** loss of cortical control of the **pontine micturition centre (PMC)** $\pm$ facilitated excitation; glutamatergic, dopaminergic and γ -aminobutyric acid (GABA)-ergic changes; $\uparrow$ cytokines and neuronal nitric oxide synthase in cord and bladder (middle cerebral artery occlusion models).

Brainstem Stroke

- Brainstem = midbrain, pons, medulla; **PMC coordinates micturition**.
- **Midbrain lesions — no LUTS; medullary and especially pontine lesions — LUTS** (nocturnal frequency, voiding difficulty, retention, UI).
- **Pontine infarct $\rightarrow$ low capacity (and, in one series, detrusor underactivity [DUA]); medullary infarct $\rightarrow$ emptying disorder.** Patterns inconsistent $\rightarrow$ **urodynamics prudent**.

Dementia

- Atrophy of grey and white matter, **especially frontal lobes**; causes include vascular, **Alzheimer (principal cause in the elderly)**, Pick, Creutzfeldt–Jakob, syphilis, traumatic brain injury (TBI), encephalitis. Prevalence **$\sim 6\%$ at ≥ 65** .
- LUTS not constant, but when present = **UI** — from DO or a **normal but inappropriately timed reflex**: patient may retain sphincter control yet void anywhere because **impaired mentation removes the reason not to**.
- **UI 3 $\times$ more common with dementia**; prevalence estimates **30–100%**, confounded by ageing and under-reporting; higher in Lewy body, Pick and normal-pressure hydrocephalus (NPH) than Alzheimer.
- **Anticholinergics** worsen cognition, delirium and falls; Alzheimer involves **cortical cholinergic loss**. Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) white paper: anticholinergic use for overactive bladder (OAB) carries **10–65% increased risk of new-onset dementia**, possibly dose-dependent $\rightarrow$ **$\beta 3$ agonist strongly preferred**.

Traumatic Brain Injury

- **Most common severe neurologic impairment from trauma**; 27–69 million/year globally.
- **Initial areflexia possible. Above the PMC $\rightarrow$ DO with coordinated sphincters; isolated brainstem injury below the PMC $\rightarrow$ DSD may be added.**
- Early UI common; retention (PVR ≥ 100 mL) less so. **UI/storage symptoms $\rightarrow$ bilateral lesions, poor function; retention $\rightarrow$ diabetes or faecal impaction.** Many remain incontinent after rehabilitation; **concomitant spinal injury** complicates management.

Brain Tumour

- Dysfunction depends on **location, not histology**; **superior frontal lobe** most implicated.
- **DO + UI**; patients may be **unaware of all LUT events** $\rightarrow$ cannot attempt suppression. Sphincters synergic; pseudodysynergia possible.
- Improvement after resection suggests tumour-related overactivity may be a **positive (activating)** rather than a release phenomenon. **Frontal space-occupying lesions can cause retention** without other deficits. **Posterior fossa tumours: voiding dysfunction in 32–70%.**

Cerebellar Ataxia

- Cerebellar degeneration ± brainstem, cord, dorsal roots: incoordination, ↓ deep tendon reflexes (DTRs), dysarthria, dysmetria, choreiform movements.
- **UI with DO and synergic sphincters**; retention/high PVR from **areflexia** or **DSD (cord involvement)**.
- **Stress UI (SUI) without DO** in some spinocerebellar degeneration → attributed to **Onuf nucleus** → **puddendal involvement**.

Normal-Pressure Hydrocephalus

- **Classic triad: dementia, gait ataxia, UI**; normal cerebrospinal fluid (CSF) pressure, dilated ventricles; idiopathic or after CVA.
- **UI in 98–100%**, from **DO with synergic sphincters**.
- **Ventriculoperitoneal shunt** is the mainstay and improves urgency and UUI.

Cerebral Palsy

- **Non-progressive brain injury** in the first year (up to 3 years); infection or hypoxia; delayed motor development, altered tone, exaggerated reflexes.
- **Most have urinary control and normal storage/emptying**. Systematic review: **~55% report ≥1 LUTS, ~46% UI**; risk ↑ with **spastic subtype, moderate–severe functional impairment, severe cognitive impairment**.
- Pattern = **suprapontine: DO with coordinated sphincters**; **DSD** in a minority (spinal damage). Voiding difficulty may reflect **failure of pelvic floor relaxation rather than true DSD**; **retention appears in adults** — difficulty may progress with age.
- **Adults with CP and LUTS: poor compliance and detrusor leak point pressure (DLPP) ≥40 cm H₂O in about half → full evaluation** (renal risk, worse with pelvic floor spasticity).
- Management: **conservative (education, pads, perineal hygiene) succeeds in ~¾**; some need intermittent catheterisation or reconstruction. **Cognitive impairment is not a barrier** to continence, but limits options needing cooperation.

Parkinson Disease

- **Pathology**: loss of dopaminergic neurons of the **substantia nigra pars compacta** (nigrostriatal tract to caudate and putamen) → **tremor, rigidity, bradykinesia**. Other contributors: striatum, thalamus, periaqueductal grey (PAG), PMC, ventral tegmental area.
- **Diagnosis: gold standard = neuropathology** — **Lewy neurites** and **Lewy bodies** (intracytoplasmic eosinophilic hyaline inclusions; not PD-specific). **Asymmetry + resting tremor + levodopa response** best separate PD from other parkinsonism (none individually specific).
- **Epidemiology**: both sexes equally; **0.3% of population, 3% at ≥65; second commonest neurodegenerative disease after Alzheimer**.
- **Other parkinsonism**: MSA, progressive supranuclear palsy (PSP), corticobasal degeneration, vascular parkinsonism, Lewy body dementia; **~5% of "PD" = Parkinson-plus** (early dementia/falls, symmetry, wide-based gait, autonomic failure; worse prognosis).
- **LUT dysfunction in 35–70%**; average **~5 years after onset**, but **LUTS at onset in up to half** — associated with **worse progression**, depression and poorer quality of life (QoL); **frequency and nocturia** most damaging.
- **Mechanism**: basal ganglia normally **inhibit** the micturition reflex; nigral cell loss removes it. Proposed: **loss of inhibitory D1 receptors allows facilitatory D2 action**; central D2 stimulation ↓ capacity and worsens DO. Early **delayed perception of filling** (visceral sensory deficit).

- **Symptoms: storage (urgency, frequency, nocturia, UUI) in 50–75%**; remainder obstructive or mixed.
- **Urodynamics: DO commonest**; smooth sphincter synergic; sporadic striated activity during involuntary contraction (in many) is **not obstructive and not true DSD**. **Pseudodyssynergia** and **sphincter bradykinesia** (delayed relaxation) mimic DSD. Impaired contractility possible; **areflexia uncommon** — suggests MSA. Positron emission tomography (PET): DO activates cerebellum most, with no pontine change.
- **Outlet surgery**: older "PD" cases with post-TURP incontinence were MSA. **TURP is not contraindicated in PD** (striated acontractility extremely rare) — but **confirm PD, exclude MSA, and do (video)urodynamics**; poorly sustained contractions ± slow sphincter relaxation lower expectations.
- **PD drugs**: levodopa, anticholinergics and dopamine agonists show no consistent LUTS effect; **bromocriptine worsens urgency and DO but improves emptying**.
- **Treatment**:
 - **Oral OAB drugs effective** (meta-analysis of randomised trials: ↓ nocturia, frequency, incontinence, ↑ QoL, no excess serious events). **Preferred (limited central nervous system [CNS] effect): mirabegron, extended-release fesoterodine, trospium, darifenacin (SUFU)**.
 - **Botulinum toxin** effective (up to 9 months); **posterior tibial nerve stimulation (PTNS)** and **sacral neuromodulation (SNM)** effective (most proceed to implant; many stop oral drugs).
 - **Subthalamic nucleus deep brain stimulation (DBS)** improves bladder sensation and capacity by **modulating afferent processing** (anterior cingulate and frontal cortex); benefit on **nocturia inconsistent** between studies.
 - **Repetitive transcranial magnetic stimulation** — capacity and first sensation improve for **~2 weeks**.
 - Limits: mobility (toilet access), **hand function for clean intermittent catheterisation (CIC)**.

Multiple System Atrophy

- **Glial α -synucleinopathy**, progressive, unknown cause: parkinsonism, cerebellar, **autonomic (urinary and erectile)** and pyramidal features. **MSA-P** (parkinsonian) and **MSA-C** (cerebellar) replace **striatonigral degeneration, olivopontocerebellar atrophy and Shy-Drager syndrome** (orthostatic hypotension, anhidrosis; possibly late-stage MSA).
- **Diffuse cell loss** (cerebellum, substantia nigra, globus pallidus, caudate, putamen, inferior olives, intermediolateral columns, Onuf nucleus) explains **earlier, more severe LUTS and erectile dysfunction (ED)** than PD. Sexes equal; **middle-age onset; poor prognosis**.
- **Early**: urgency, frequency, UUI **up to 4 years before diagnosis**; DO, ↓ **compliance** (distal cord autonomic neurons). **Late**: difficulty initiating/maintaining voiding (pontine and sacral lesions) — poor prognosis.
- **Outlet: open bladder neck** on cystourethrography and **striated sphincter denervation** on motor-unit EMG → sphincteric UI in women; **prostatectomy hazardous** (most MSA men worsen after TURP; women incontinent after pelvic floor repair).
- **Treatment difficult**: treating DO worsens emptying; treating sphincteric UI worsens emptying; **outlet reduction rarely indicated**. Goal = **facilitate storage + CIC** (often impossible late). **Desmopressin helps predominant nocturia**; antimuscarinics usually disappoint.

BOX 109.1 UROGENITAL CRITERIA FAVOURING DIAGNOSIS OF MULTIPLE SYSTEM ATROPHY — MSA VS PARKINSON DISEASE

Feature	MSA	Parkinson disease
Timing of urinary symptoms	Precede or accompany parkinsonism	Years after diagnosis

Feature	MSA	Parkinson disease
ED	Precedes or accompanies parkinsonism ; near universal	$\frac{1}{3}$; follows diagnosis by 1–4 years
Dominant urinary problem	UI	Frequency/urgency; UI less common
PVR	High	Usually normal
Bladder neck	Open	Synergic
Striated sphincter EMG	Denervation	Normal (bradykinesia only)
Detrusor	DO → areflexia	DO; areflexia uncommon
After urologic surgery	Worsens	TURP acceptable after work-up
Response to treatment	Poor; desmopressin for nocturia	Antimuscarinics, β_3 agonist, toxin, SNM effective

- Box 109.1 lists the first seven criteria (bolded MSA column); the remaining rows are the text contrasts.

Progressive Supranuclear Palsy

- Parkinsonism + **vertical gaze palsy**, falls, cognitive impairment. LUTS: frequency, urgency, UI.
- **Impaired detrusor contraction very common, DO common**; sphincter abnormalities reflect **Onuf nucleus neuronal loss**.

Concept — *above the pons the reflex is intact but disinhibited, so the expected pattern is DO with synergic sphincters; true DSD needs a lesion below the PMC, and any open bladder neck or sphincter denervation points to a multisystem (autonomic) process such as MSA.*

DISEASES PRIMARILY INVOLVING THE SPINAL CORD

Multiple Sclerosis

- **Epidemiology**: adults 20–50 (onset commonly 30–38), **female predominance ($\geq 2:1$; ~75% female)**; **2.9 million worldwide**; a leading cause of neurogenic LUT dysfunction (NLUTD). Optic, pyramidal, ataxic, bladder, bowel and sexual symptoms; **progressive forms are predominantly spinal**.
- **Pathology**: immune-mediated **demyelination with axonal sparing** → impaired saltatory conduction; **plaques** in white matter; commonest in **lateral corticospinal and reticulospinal columns of the cervical cord** (cervical almost always; lumbar ~40%, sacral ~18%). Cortical/midbrain lesions → intellectual change/euphoria.
- **Functional imaging**: neurogenic DO (NDO) activates frontal executive areas (inhibitory signal); DSD adds emotional, motor and brainstem activation. Mild-obstruction/preserved-contraction patients have **denser calcitonin gene-related peptide (CGRP) and substance P innervation**.
- **Prevalence**: LUTS in 50–90%, UI 37–72%, retention 2–52%; LUT dysfunction **correlates with disability**; **isolated presenting complaint in up to 15%** (acute retention or acute urgency/frequency); up to 80% affected at some point.
- **Urodynamics**: **DO commonest (34–99%)**; **DSD with DO in 30–65%**; **impaired contractility/areflexia 12–38%** (complicates treatment); **~10% normal**; smooth sphincter synergic; sphincteric flaccidity <15%; pelvic floor weak; **pelvic floor spasticity only with DSD**. Multiple lesions limit diagnosis from urodynamics alone.

COMPARISON — AVERAGE URODYNAMIC PATTERNS IN MS (REVIEW OF SERIES)

Pattern	Average
DO + synergic striated sphincter	~38%

Pattern	Average
DO + striated DSD	~29%
Detrusor areflexia	~26%

- **Sensation usually intact → distinguish pseudodyssynergia from true DSD.** Brief DSD in a woman may be harmless; **sustained DSD with prolonged high pressure causes complications**; contraction amplitude relates to upper tract deterioration.
- **Risk factors for complications: (1) DSD in men; (2) high detrusor filling pressure; (3) indwelling catheter.**
- **Upper tracts:** international committee — progressive MS **rarely damages the upper tracts**, unlike SCI; others find disability predicts poor compliance and vesicoureteral reflux (VUR); **VUR rare (~5%), linked to progressive MS. No factor predicts upper tract change**; symptoms do not track disease → **baseline urodynamics essential**.
- **Management:**
 - **Aggressive, anticipatory medical management** — surgery needed in <10%, none developed hydronephrosis (regimens: antimuscarinic + CIC commonest; drugs alone; CIC alone; behavioural).
 - **Avoid irreversible therapy** — urodynamic pattern and compliance change over time; surgery declining.
 - **Individualised bladder rehabilitation** (re-education, pelvic floor, emptying techniques, bowel programme) improves QoL in a randomised trial.
 - **Early MS: antimuscarinics + CIC are the cornerstone.** Advanced MS (Expanded Disability Status Scale [EDSS] ≥7): no specific guidelines.
 - **Credé/Valsalva contraindicated, especially with DSD.**
 - **Indwelling catheter only after all else fails; if used, suprapubic** (~30% use indwelling).
 - **OnabotulinumtoxinA:** durable, safe; majority achieve ≥50% UI reduction (MS and SCI).
 - **SNM** effective in small series; **MRI-compatible devices** have widened eligibility.

Concept — MS is **multifocal and relapsing**, so the pattern (DO, DSD, areflexia) is unpredictable and changes; hence **baseline urodynamics, reversible therapy and repeated reassessment**, and a lower upper-tract risk than SCI.

Spinal Cord Injury

Epidemiology, Morbidity, General Concepts

- **Causes:** violence, fracture/dislocation (motor vehicle, diving, falls), vascular injury or surgery, infection, disc prolapse, hyperextension.
- **Risks:** UTI, sepsis, upper and lower tract deterioration and calculi, autonomic dysreflexia, skin breakdown, psychological morbidity.
- **Anatomy:** column (vertebral) and cord levels diverge — **always state which. Sacral cord begins at column T12-L1; cord ends (conus) → cauda equina at ~L2.** Injury may extend cephalad/caudad; multiple levels occur. Complete anatomic transection is rare.
- **US data:** ~54 per million (~17,000 new cases)/year; ~302,000 prevalent. Mechanisms: **motor vehicle > falls > violence > sport**; **80% male**; mean age at injury risen from **29 (1970s) to 42**. **Incomplete tetraplegia largest group**; **<1% full recovery by discharge**. Most injuries at/above T12 column; cervical → tetraplegia, thoracic/lumbar/sacral → paraplegia.
- **Long-term:** UI persists in many; **paraplegics leak more often than tetraplegics** (catheter dependence); CIC associated with more UI than other methods.

- **Mortality shift:** renal failure was the leading cause (pre-World War II and earlier data); now **pneumonia and septicemia** — reflects better bladder care.
- **Stones:** ~7% within 10 years, **greatest risk in first 3 months; 98% apatite or struvite**; two peaks — **acute (immobilisation hypercalciuria)** and **chronic (catheter-related, mainly lower tract)**.
- **Experimental:** contusion level alters retention (T1 lesions spare a descending pathway); crush worse than contusion; **KATP channels regulate spontaneous activity in neurogenic muscle, BKCa in normal**.
- Stem-cell repair, reinnervation and sexual/reproductive function are outside this chapter.

KEY POINTS: SPINAL CORD INJURY

- **Complete SCI above the sacral cord but below the sympathetic outflow → DO, absent sensation below the lesion, smooth sphincter synergia, striated sphincter dyssynergia.**
- **Lesions at or above cord T7–T8 (column T6) may add smooth sphincter dyssynergia.**
- **Neurologic examination is no substitute for urodynamics in defining risk and treatment.**

Spinal Shock

- **Decreased excitability of cord segments at and below the lesion:** absent somatic reflexes, flaccid paralysis; autonomic activity also suppressed → **acontractile, areflexic bladder**, smooth contour, no trabeculation.
- **Sacral reflexes (anal, bulbocavernosus) may never disappear, or return within minutes-hours;** function above the lesion may also be depressed. Pathophysiology poorly understood (grey matter trauma, protein accumulation, reperfusion).
- **Outlet: bladder neck closed and competent** (unless prior surgery or thoracolumbar sympathetic injury); some striated EMG activity; **maximum urethral closure pressure (MUCP) reduced but maintained; guarding reflex absent, no voluntary control → UI only with overflow.**
- **Management: retention is the rule — catheterise; keep storage pressure low. CIC preferred,** though small-bore urethral or suprapubic catheters give similar outcomes at this stage.
- **Recovery:** if the distal cord is intact, **reflex contractions return** — weak and poorly sustained at first, then stronger → **involuntary voiding between catheterisations, alongside return of lower-limb DTRs.**
- **Duration:** usually 6–12 weeks in complete suprasacral lesions; up to 1–2 years; shorter if incomplete.

Suprasacral Spinal Cord Injury

- **Mechanisms of reflex recovery (C-fibre-evoked):** loss of bulbospinal inhibition; synaptic strengthening/sprouting; transmitter changes; **afferent sprouting**. Animal findings: **C-fibre sensitisation (NGF)**, dorsal root ganglion enlargement, **sodium-channel shift from high-threshold tetrodotoxin-resistant to low-threshold tetrodotoxin-sensitive**; ↑ glutamate/glycine/taurine; urothelial barrier disruption; **prejunctional M1 → M3 shift**; ↑ urothelial ATP release (nicotinic/purinerbic, atropine-resistant); ↑ spinal NGF; myosin heavy-chain changes; **TRPA1** modulation attenuates overactivity.
- **Characteristic pattern: DO + smooth sphincter synergia (if below sympathetic outflow) + striated DSD → incomplete emptying.** Examination: spasticity, hyperreflexia, abnormal plantars, impaired sensation. **Guarding reflex absent/weak if complete, preserved but variable if incomplete.**
- **DSD timing:** external sphincter contraction **starts before the detrusor contraction** in most (Delay A ~2 s; Delay B ~7.6 s to +10 cm H₂O) — associated with complete injury and continuous DSD.
- **DSD → functional obstruction, poor emptying, high pressures.** Poorly sustained contraction may occur near the **conus** (occult second lesion, pelvic-floor inhibitory reflex, loss of higher facilitation). **Reflex voiding can be triggered** (suprapubic tapping).

- **Severity of DSD consequences: complete > incomplete; continuous > intermittent; male > female.**
- **DSD type vs lesion: incomplete → type 1; complete → types 2 and 3; no relation to level; type evolves in about a quarter, stable in ~2/3.**
- **Management logic: it is both storage and emptying failure. If pressures are (or can be made) low → treat as emptying failure with CIC.**
 - **Antimuscarinics, β3 agonists or both** (preponderance of evidence); **intradetrusor botulinum toxin** (randomised trials) if refractory or lifelong drugs unsuitable.
 - **Sphincterotomy or intrasphincteric onabotulinumtoxinA** (men) lowers DLPP and converts to **storage failure** managed with a condom collector — **only for those unwilling/unable to do CIC**; counsel on failure, repeat procedures, monitoring, skin breakdown.
 - **Anterior sacral root stimulation with deafferentation** established; ventral root rerouting and stem cells investigational.

KEY POINTS: MANAGEMENT OF PATIENTS WITH SUPRASACRAL SPINAL CORD INJURY

- **Pattern: DO, smooth sphincter synergia (below sympathetic outflow), striated sphincter dyssynergia.**
- **Initial and periodic evaluation to find and correct: high-pressure storage, high DLPP, VUR, stones (upper and lower), infection — especially with reflux.**
- **CIC for emptying failure; oral drugs (anticholinergic, β3 agonist) or chemodenervation for storage failure.**

Sacral Spinal Cord Injury

- After shock: ↓ DTRs, flaccid paralysis, absent sensation. **Detrusor areflexia with normal or high compliance initially; ↓ compliance may develop** (decentralisation-induced plasticity).
- **Outlet: competent but non-relaxing smooth sphincter; striated sphincter with fixed tone, no voluntary control; closure pressures ↓. Bladder neck may later open.**
- **Straining/Credé → "obstruction"** at a closed bladder neck or fixed distal sphincter. Fig. 109.4: **low compliance, incompetent bladder neck, distal sphincter fails to open on straining** — typical of sacral cord or efferent root injury.
- **Management: CIC for emptying + keep storage pressure acceptable (oral drugs or botulinum toxin).**

KEY POINTS: MANAGEMENT OF PATIENTS WITH SACRAL SPINAL CORD INJURY

- Same risk factors, with emphasis on **storage pressure** — **silent upper tract deterioration can occur without VUR.**
- **Low-pressure storage + CIC**; pharmacologic or electrical stimulation may aid emptying.

COMPARISON — SUPRASACRAL VS SACRAL SCI (AFTER SPINAL SHOCK)

	Suprasacral (complete)	Sacral (complete)
Detrusor	Overactive	Areflexic
Compliance	Normal initially	Normal/high → may fall
Smooth sphincter	Synergic (dyssynergic if above T7–T8)	Competent non-relaxing → may open
Striated sphincter	Dyssynergic	Fixed tone , no voluntary control
Somatic signs	Spasticity, hyperreflexia	Flaccidity, ↓ DTRs
Failure mode	Storage + emptying (high pressure)	Emptying ± low-compliance storage
Management	Lower pressure (drugs/toxin) + CIC	CIC + keep compliance safe

Neurologic and Urodynamic Correlation

- **Correlation is general, not exact:** somatic completeness $\neq$ autonomic completeness; multiple occult levels.
- Incomplete SCI: bladders "not at risk" deteriorate and vice versa; **management changes are often needed without neurologic change.**
- **Classic correlation (550 patients):** suprasacral lesions $\rightarrow$ normal voiding, DSD, or — paradoxically — **areflexia** (occult distal lesion or disordered sacral integration). **DSD occurred only with suprasacral cord lesions, never with supraspinal lesions $\rightarrow$ (1) coordination is supraspinal; (2) DSD implies a lesion between the pontine-mesencephalic reticular formation and the sacral cord.** Suprapontine patients voided synergically; many with DO kept voluntary sphincter control (**separate pathways for bladder and sphincter**); voluntary sphincter contraction did **not** abolish detrusor contraction (complex, not a simple sacral reflex). Mixed patterns reflect **separate parasympathetic and pudendal nuclei.**
- **Thoracolumbar (column T10–L2) lesions least predictable** — roughly half areflexic, half DO.
- **With precise imaging (CT/MRI):** suprasacral $\rightarrow$ **DO and/or DSD in 95%**, low compliance (<12.5 mL/cm H₂O) and DLPP ≥ 40 cm H₂O in $\sim 40\%$; sacral $\rightarrow$ **areflexia, low compliance and high leak point pressure in most**; combined lesions intermediate.
- **Upper tract risk:** detrusor **fibrosis** severity; **leak point pressure ≥ 75 cm H₂O; DO for more than one-third of filling $\rightarrow$ renal damage $\rightarrow$ treat NDO aggressively.**
- **Repeat urodynamics** vary in capacity/pressures but reliably detect DO — one study may not suffice.
- **Bottom line: manage by urodynamics, not neurology; do not infer neurology from urodynamics** (complete-lesion data largely valid). Regular urodynamic follow-up.

Autonomic Dysreflexia

- **First described by Guttman and Whitteridge, 1947. Potentially fatal emergency unique to SCI above cord T6–T8 (sympathetic outflow); cervical (60%) > thoracic (20%); onset usually after spinal shock, possibly years later; requires viable distal cord.**
- **Features:** exaggerated sympathetic response to stimuli **below** the lesion — **pounding headache, hypertension, flushing and sweating above the lesion; bradycardia typical** (tachycardia/arrhythmia possible). Range: mild pre-void headache $\rightarrow$ **cerebral haemorrhage, seizure.**
- **Triggers: bladder or rectal distension** (instrumentation, tube change, blocked catheter, clot); calculi, gut pathology, long-bone fracture, sex, electrocoagulation, pressure sores, sport. **Resolves quickly when the stimulus is removed.**
- **Urodynamic risk factors:** complete injury, cervical level, DSD, poor compliance, ≥ 2 years since injury. **DSD invariable; smooth sphincter dyssynergia usual in men.**
- **Mechanism:** afferent nociceptive input $\rightarrow$ spinal reflex sympathetic outflow (arteriolar, pilomotor, visceral spasm, sweating) **without descending medullary inhibition**; loss of bulbospinal control of preganglionic neurons; large vascular bed (probably **splanchnic**) involved; sympathetic plasticity.
- **During urodynamics** (defined as systolic rise ≥ 20 mm Hg): common at $\geq T6$, **but also occurs below T6**; associated with continuous DSD and poor compliance; **bradycardia in a minority $\rightarrow$ monitor all SCI patients.**
- **Prevention:** endoscopy under **spinal or carefully monitored general anaesthesia**; urodynamics with **continuous haemodynamic monitoring.**
- **Treatment sequence:** remove stimulus (drain bladder) $\rightarrow$ drug therapy if persistent $\rightarrow$ neuroablation if intractable.

COMPARISON — DRUGS FOR AUTONOMIC DYSREFLEXIA

Agent	Class / mechanism	Role	Note
Nitroglycerin 2% paste	Nitrate → nitric oxide → vasodilatation	Current first-line acute therapy	Applied above the lesion
Sodium nitroprusside	Intravenous nitrate	Persistent hypertension despite nitrates	
Nifedipine	Calcium antagonist	Sublingual 10–20 mg during cystoscopy; 10 mg orally 30 min before	Banned in many centres — uncontrolled, irreversible hypotension
Labetalol	α/β blocker	Rapid-acting; favoured by anaesthetists	
Captopril, hydralazine, diazoxide	Vasodilators	Occasionally used	Less advantageous
Ganglionic blockers	Ganglion blockade	Former mainstay	Abandoned
Terazosin (nightly, titrated 1–10 mg)	Selective α_1 blocker	Prophylaxis — reduces severity	Erections and BP unchanged; dizziness
Prazosin	α_1 blocker	Prophylaxis	

- **Prophylaxis matters** because **silent BP surges** occur; it does **not** remove the need for monitoring.
- **Intractable cases:** sympathectomy, sacral neurectomy, **sacral rhizotomy** (moderately invasive, low risk, combined with intermittent catheterisation), cordectomy, dorsal root ganglionectomy.

Vesicoureteral Reflux

- **17–25% of SCI**, more in suprasacral; drivers = **high intravesical pressure (filling and voiding) and infection**. Persistent VUR → chronic renal damage (common in SCI deaths from renal disease); **high pressures damage kidneys even without VUR**.
- **Chronic urethral catheterisation is worst for the upper tracts** (more hydronephrosis than CIC or spontaneous voiding).
- **First treat the bladder, not the reflux: lower storage pressure and outlet resistance** — drugs, urethral dilatation (myelomeningocele), neuromodulation, deafferentation, augmentation, sphincterotomy.
- Failing that, anti-reflux surgery in a thick bladder is hard; **transureteroureterostomy** complicated by stones, recurrent VUR, obstruction; **submucosal bulking** adds an option.
- **Destroyed urethra** (long-term catheter): **bladder neck closure + suprapubic catheter**; diversion (including detubularised segments) remains common.
- **Urodynamic artefact:** VUR makes capacity look larger, pressures lower and DO less significant.

Spinal Cord Injury in Females

- ↓ protein and bone mass, ↑ fat → skin breakdown, fractures, osteoporosis; **menopausal flushes mimic autonomic dysreflexia**; UI and UTI worsen with age.
- **Paraplegic women usually master CIC; lack of an external collecting device is the core problem**. Most tetraplegic women need indwelling catheters (leakage around catheter, upper tract change).
- **Complications:** CIC $\ll$ reflex voiding with pads < indwelling catheter. Poor-recovery cervical SCI in women with permanent catheters = frequent **bladder stones, bypass leakage, recurrent infection** → viewed as **management failures**; good recovery → most void by triggering.
- **SUI surgery in selected women:** mid-urethral or **autologous slings** reasonable; bulking less invasive but weak data.

Cervical Myelopathy

- Compression from **spondylosis, ossification of the posterior longitudinal ligament, disc herniation**.
- Storage and obstructive symptoms; **DO common, DSD in some — yet underactive detrusor in others**, unexpected for cervical disease → **urodynamics guide therapy**.

Acute Transverse Myelitis

- **Rapid motor, sensory and sphincter loss with a defined upper sensory level, no compression**; parainfectious, autoimmune, vascular or demyelinating; **stabilises in 2–4 weeks**, non-progressive, variable recovery.
- **Retention at presentation almost universal**; acute phase often areflexic; after shock **DO predominates, DSD common**; persistent retention or storage symptoms common at 6 months; **severity does not predict urodynamics**.
- Long-term: many need antimuscarinics and **intradetrusor toxin**, a few augmentation; some develop hydronephrosis → **urodynamics before irreversible therapy; manage as SCI**.

Neurospinal Dysraphism

- **Definitions: spina bifida occulta** = bony arch defect only; **spina bifida cystica (aperta)** = bony + neural tube defect: **myelomeningocele** (neural elements herniated; **~90% of cystica**, most devastating) vs **meningocele** (meningeal sac only); prefix **lipo-** if fat present.
- **Level**: cervical 2%, thoracic 5%, lumbar 26%, **lumbosacral 47%**, sacral 20% — **level correlates poorly with urodynamics**.
- **Incidence ~1/1000 live births; LUT dysfunction ~90%**; falling with **folate** and prenatal diagnosis; **up to 85% reach adulthood** → transitional care.
- **"Typical" adult myelodysplastic bladder: areflexic with an open bladder neck**; fills until **fixed external sphincter pressure** is reached, then leaks; SUI with abdominal pressure. **DSD in 10–15%** — these have a **functional bladder neck** and can be continent if the reflex is controlled. Continence often **improves after puberty**, but tolerance of leakage falls.
- **Neurologic examination fails most obviously here**: mix of DO/areflexia, **low compliance common**, DSD, low urethral pressure, **striated denervation in the majority**.
- **Main goal: avoid high storage pressure and upper tract deterioration**. Poor compliance, high voiding pressures and DO predict renal decline; childhood DO predicts later bilateral renal dysfunction.
- **SUI**: raise outlet efficiency **without raising closure pressure enough to reduce compliance** — **periurethral bulking safer than pubovaginal sling or artificial urinary sphincter**; men: bulking ± "prostatic sling"; continent patients stay on CIC.
- **Management**: medical + **intravesical onabotulinumtoxinA**, then **augmentation or diversion** on failure; surgery **not necessarily superior** (higher UI in surgical cohort — confounded by severity). Augmentation patients younger and more often male; conduit patients older, female, costlier.
- **Xiao procedure** (L5 → S3 ventral root anastomosis): with detethering ↑ capacity and ↓ overactivity, **no continence gain** (small randomised trial).
- **Tethered cord syndrome (TCS): stretch-induced dysfunction of a caudally anchored cord** (scar, thick filum, bony septum, tumour); **3–15% of myelomeningocele**. Adults: prior dysraphism with late progression, or new subtle symptoms; **children present after growth spurts, adults after spinal stretch (sport, collision)**. Symptoms: back pain, leg weakness, foot deformity, scoliosis, sensory loss, bowel/bladder dysfunction. **No typical pattern — urodynamics; DO most frequent. Release improves**

urodynamics in a minority; urinary recovery lags (~4 months) behind pain (~1) and motor (~2); most improvement within 6–12 months.

- **Transition to adult care:** multidisciplinary; goals = **low detrusor pressure, preserved compliance, social continence; lifelong follow-up mandatory.**

Tabes Dorsalis, Pernicious Anaemia

- **Tabes (syphilitic myelopathy):** dorsal columns and posterior sacral roots → **loss of bladder sensation, large PVR = classic "sensory neurogenic bladder";** some show low compliance or DO.
- **Pernicious anaemia:** vitamin B12 malabsorption (no intrinsic factor) → **subacute combined degeneration** → same sensory pattern.

Subacute Combined Degeneration

- **Demyelination of posterior and lateral columns; B12 deficiency** (diet, malabsorption, gastrectomy, fish tapeworm, autoimmune) or **copper deficiency.**
- Mixed storage and voiding LUTS; areflexia, NDO or normal; **B12 replacement reverses dysfunction** in many.

Poliomyelitis

- When present: **"motor neurogenic bladder" — retention, areflexia, intact sensation;** LUT dysfunction reported in 4–42%; survivors have more hesitancy, weak stream, incomplete emptying, nocturia, UII.

COMPARISON — SENSORY VS MOTOR NEUROGENIC BLADDER

	Sensory	Motor
Lesion	Dorsal columns / posterior roots	Anterior horn / motor efferents
Examples	Tabes, pernicious anaemia/SACD, diabetes	Poliomyelitis
Sensation	Lost	Intact
Detrusor	Decompensates from overdistension	Areflexic
Capacity	Large	Retention

Concept — in the cord, level decides pattern — suprasacral lesions give *high-pressure DO with DSD*, sacral lesions *areflexia with fixed outlet* — but *completeness, multiple levels and plasticity* break the rule so often that *urodynamics, not the neurologic level, directs management*, and the enemy throughout is *sustained high storage pressure*.

DISEASE DISTAL TO THE SPINAL CORD

Disc Disease

- **Anatomy:** sacral cord segments lie at **L1–L2 vertebral bodies (conus);** sacral roots descend behind L2–L5 as the **cauda equina** to exit at sacral foramina — so **any lumbar prolapse can compress sacral roots.**
- **Most prolapse is posterolateral (spares the cauda equina);** central prolapse in 1–15% → **compression.** Commonest levels **L4–L5, L5–S1.**
- **Signs:** low back pain radiating in a girdle; **perineal/perianal sensory loss (S2–S4), lateral foot sensory loss (S1–S2).**
- **Incidence 27–68% (biased series). Commonest finding: normally compliant areflexic bladder** with normal or partially denervated pelvic floor; patients **strain to void** or present in retention; **DO** occasionally from root irritation.

- **Compliance usually preserved (unlike myelomeningocele)** — because disc injury is **more incomplete for preganglionic parasympathetic fibres** and **more sensory than motor**; low compliance in myelomeningocele is attributed to **preganglionic parasympathetic motor injury**.
- **Laminectomy may not restore function**; some develop new DO or areflexia after surgery → **pre-operative urodynamics** (clinical and medicolegal).
- **Cauda equina syndrome: perineal sensory loss + loss of voluntary anal and urethral sphincter control + loss of sexual response**; 1–5% of prolapsed lumbar discs; initially acontractile detrusor, no sensation, sphincter EMG often silent. **Somatic (sphincter EMG) recovers before autonomic (detrusor)**.
- **Urgent decompression — within 48 hours** gives significantly better urinary and rectal recovery than later surgery (meta-analysis); most regain continence, but recovery is variable.

Spinal Stenosis

- Narrowing of canal, root canals or foramina (congenital, developmental, acquired), **may occur without disc prolapse**; compression → damage, **ischaemia, oedema**.
- **Lumbar stenosis**: exercise-related back/leg pain, cramping, paraesthesia **relieved by rest** (sacral root ischaemia). Spectrum from cervical cord compression to cauda equina syndrome.
- **Cervical spondylotic stenosis: DO if inhibitory reticulospinal tracts compressed; underactivity if posterior funiculus (proprioception) involved**.
- No consistent pattern → **urodynamics are the cornerstone; decompression improves bladder symptoms in about half** and improves PVR, capacity and flow after successful surgery.

Radical Pelvic Surgery

- **Pelvic (inferior hypogastric) plexus** lies beside the rectum (and vagina); injury commonest after **abdominoperineal resection (APR) and radical hysterectomy**.

Procedure	Historical incidence of LUT dysfunction
APR	20–68%
Radical hysterectomy	16–80%
Anterior resection	20–25%
Proctocolectomy	10–20%

- **Current rates lower with nerve-sparing; permanent in 15–20%** of those affected. Rectal cancer treatment: ~ $\frac{1}{3}$ have LUT dysfunction; **APR has greatest impact**; chemoradiotherapy compounds it; sexual dysfunction common.
- **Mechanisms**: denervation/decentralisation, nerve tethering or scar, direct trauma, devascularisation.
- **Decentralisation plasticity**: parasympathetic decentralisation → **adrenergic hyperinnervation**, converting sympathetic **β (relaxant) bladder response to α (contractile)**; not seen with pelvic plexus neurectomy alone or with added hypogastric neurectomy; new cholinergic inputs from hypogastric nerves; **urethral α -adrenergic supersensitivity**.
- **Permanent pattern: impaired/absent contractility + fixed residual striated tone (not voluntarily relaxed) + open, non-functional smooth sphincter + ↓ compliance → storage and emptying failure**: leak across the distal sphincter with abdominal pressure (**obvious in women; prostate masks it in men**), or retention. **Positive bethanechol supersensitivity test**.
- **Therapeutic goal: low-pressure storage + periodic emptying**.
- **Avoid prostatectomy unless BOO is proven at the prostate** — otherwise it only weakens the sphincter.

- **Early post-operative period: CIC and wait — most dysfunction is transient; 6–12 months may pass before detrusor recovery.** Decreased compliance is more frequent after radical pelvic surgery than after conus/cauda equina injury.

Simple and Radical Hysterectomy

- **Simple hysterectomy:** no consensus on neurogenic damage; total vs subtotal — **less UI after total**, both ↑ incomplete emptying (double voiding); vaginal route associated with more micturition and defecation symptoms than abdominal.
- **Radical hysterectomy:** potentially more debilitating, though one survey found no excess long-term dysfunction; **nerve-sparing radical hysterectomy reduces long-term urinary symptoms without oncological compromise (randomised trial); minimally invasive approaches** lower long-term bladder dysfunction.

Childbirth

- **No prospective long-term data; no trial shows caesarean section protects against UI.**
- **EPINCONT** (Epidemiology of Incontinence in the County of Nord-Trøndelag): **any UI higher after caesarean than nulliparity; moderate–severe UI lower after caesarean than vaginal delivery; UUI equal across groups; SUI linked to mode of delivery.**
- **Pregnancy itself** raises UI and faecal incontinence risk regardless of mode.

Herpesvirus Infections

- **Herpes zoster** invading **sacral dorsal root ganglia/posterior roots** → **retention with areflexia days–weeks after the rash** (may precede rash with fever, perineal paraesthesia, obstipation). DO/UI also possible (root irritation, meningitis/myelitis, "zoster cystitis" — bladder vesicles at cystoscopy). **Resolves spontaneously in 1–2 months.**
- Voiding dysfunction ~4% of zoster admissions, rising to **~29% with lumbosacral dermatome involvement**. Three types: **herpetic cystitis (commonest)**, neuritis → **flaccid bladder retention**, myelitis → **DO**; all normal within 8 weeks.
- **Herpes simplex (anogenital):** retention **2–7 days after genital eruption**; with **CSF pleocytosis + sacral radiculitis + sphincteric dysfunction = Elsberg syndrome**; transient (may leave sacral neuropathic pain).

Diabetes Mellitus

- **Commonest cause of peripheral neuropathy** in Europe/North America; **11.3% of US population** (28.5 million diagnosed adults). LUTS in 5–59% — distinguish referred from unselected cohorts. Nurses' Health Study: **UI more prevalent in diabetic women (odds ↑ ~20%), predominantly UUI; no SUI association.**
- **Classic diabetic cystopathy = sensory + motor neuropathy: impaired sensation** → ↑ capacity → ↓ contractility → impaired flow → later ↑ PVR.
- **Differential in men = BOO** (both give low flow) → **pressure–flow studies separate them. Smooth/striated dyssynergia not a feature** — false positives from abdominal straining (pseudodyssynergia; straining cannot open the bladder neck).
- **DO is also common** in diabetic series (often with coexisting cerebrovascular disease) → **urodynamics before treatment.**
- **Mechanisms (five sources):**
 1. **Muscarinic receptor change — M2 overexpression** (M2 preserves contraction when M3 lost).
 2. **Advanced glycation end products** → microvascular disease, endothelial dysfunction, ↓ nitric oxide.

3. **Sorbitol (polyol, aldose reductase pathway)** → ↓ myo-inositol, ↓ Na⁺/K⁺-ATPase, **reactive oxygen species/oxidative stress**.
 4. **Diacylglycerol** → **protein kinase C** → connective tissue (type IV collagen, fibronectin), neural and endothelial damage.
 5. ↓ **NGF**.
- **Further findings:** Aδ- and C-fibre dysfunction impairs emptying; autonomic neuropathy associated with LUTS; ↓ nitric oxide-mediated urethral relaxation; ↑ muscarinic and purinergic expression; ↓ sodium-pump activity; mitochondrial and gap-junction abnormalities, collagen deposition (**time-dependent**); **polyuria (diuresis) itself enlarges the bladder**; anti-voltage-gated calcium channel antibodies (type 1 model); L-type channels and TRPV1. **Animal model matters** (alloxan → DO; sucrose-fed → normal; streptozotocin → pyuria).
 - **Management:** **early timed voiding** prevents distension-induced decompensation; intensive glycaemic control slows neuropathy experimentally, but **LUTS in type 1 diabetes respond poorly to glycaemic control**.

KEY POINT: DIABETES

- **DO has been cited as the most frequent urodynamic finding** in diabetic cystopathy — possibly reflecting **timing of diagnosis within the natural history** (DO early, hypocontractility late).

Guillain-Barré Syndrome

- **Acute inflammatory demyelinating disorder of peripheral somatic and autonomic nerves: rapid symmetric weakness, areflexia**, mild sensory signs; post-infectious molecular mimicry (**two-thirds preceded by respiratory or gastrointestinal infection**). **Demyelinating form 85%, axonal 15%**. Nadir within 2 weeks (75%) / 4 weeks (94%); mortality 5–8%.
- Autonomic complications: arrhythmia, BP lability, bowel, bladder, sexual dysfunction. **LUT dysfunction 25–80%; retention 11–30%**.
- Findings: ↑ **PVR commonest**, then ↓ sensation, DO, ↓ compliance, DUA, DSD; **severity tracks motor grade/disability, not antibody titres**.
- **Manage reversibly (CIC, anticholinergics) pending recovery**.

COMPARISON — HERPES ZOSTER VS HERPES SIMPLEX VS GUILLAIN-BARRÉ

	Herpes zoster	Herpes simplex	Guillain-Barré
Site	Sacral dorsal root ganglia	Sacral roots ± CNS	Peripheral nerves (demyelination)
Onset	Days-weeks after rash	2–7 days after genital eruption	With ascending weakness
Usual LUT finding	Retention, areflexia; zoster cystitis	Retention	↑ PVR, retention
Eponym	—	Elsberg syndrome (with CSF pleocytosis)	—
Course	Resolves in 1–2 months	Transient	Reversible therapy pending recovery

Concept — *distal to the cord the lesion hits the **peripheral reflex arc**, so the signature is **areflexia with a non-relaxing or fixed outlet**, and **compliance falls only when preganglionic parasympathetic motor fibres are damaged** — which is why urgent decompression, waiting and CIC outrank outlet surgery.*

MISCELLANEOUS NEUROLOGIC DISEASES CAUSING LOWER URINARY TRACT DYSFUNCTION

COMPARISON — MISCELLANEOUS NEUROLOGIC DISEASES

Disease	Nature	LUT prevalence	Dominant urodynamic pattern	Point to remember
Amyotrophic lateral sclerosis (Lou Gehrig)	Fatal upper + lower motor neuron degeneration	LUTS ~44–50%; OAB ~¼	DO with non-relaxing external sphincter or bladder neck	Not classically linked but common — ask and treat
Lyme disease (neuroborreliosis)	Encephalopathy, polyneuropathy, leucoencephalitis; spirochaete may invade bladder	~35%	DO; areflexia in some; no DSD	Retention can be the presenting feature
Hereditary spastic paraplegia	Usually autosomal dominant; central demyelination, progressive leg spasticity	High (to ~77% in Eastern European descent)	DO, DSD, ↑ PVR	Urgency dominant; low upper tract risk ; screen systematically
Tropical spastic paraparesis	HTLV-1 myelopathy	Up to 60%	DSD and DO	Distinguish from MS
AIDS	HIV: central + peripheral nerves	Variable	DO, DSD; areflexia with CMV polyradiculopathy	Neurogenic LUTS herald poor prognosis (~40% dead within 8 months) ; rare, late-stage; do not miss BPH/OAB
Acute disseminated encephalomyelitis	Post-infectious multifocal demyelination	~⅓	DO or DUA	Retention commonest ; usually recovers
Syringomyelia	Cervical syrinx: dissociated sensory loss, brachial amyotrophy	9–25%	DO, DSD, areflexia; sphincter denervation	May improve after syringosubarachnoid shunt
Schistosomal myelopathy	Granulomatous mass or transverse myelitis	—	DO + DSD ; areflexia	DSD → upper tract decompensation ; bladder infiltration also obstructs
Systemic lupus erythematosus	Autoimmune vasculitis; myelopathy 1–3%	—	DO, impaired contractility, DSD	Lupus cystitis : thick inflamed wall, haematuria, hydronephrosis/VUR → steroids/immunosuppression
Reflex sympathetic dystrophy	Post-traumatic pain + vasomotor change	Unknown	DO or areflexia	Association real, causation unclear
Tuberculosis (spinal, meningitis)	Cord compression / meningitis	Spinal TB ~½	DO, areflexia, DSD, ↓ sensation	Bladder dysfunction predicts poor recovery ; many resolve with treatment

- **Pattern across rows: myelopathies (HTLV-1, schistosomal, HSP, syringomyelia) produce the suprasacral DO + DSD pattern; root/peripheral disease (CMV, Lyme) produces areflexia** — the lesion site, not the disease label, predicts the bladder.

MISCELLANEOUS CONDITIONS DEFINITELY, PROBABLY, OR POSSIBLY RELATED TO NEUROMUSCULAR DYSFUNCTION

Detrusor Sphincter Dyssynergia

- **Dyssynergia** = kinesiological dissociation of normally harmonious muscles; **DSD (DESD)** = **striated (external) sphincter** unless stated. Smooth sphincter = bladder neck and proximal urethral smooth muscle.
- **True DSD requires a lesion between the sacral cord and the brainstem PMC** — traumatic SCI, MS, transverse myelitis. **Without such a lesion, doubt the diagnosis and search exhaustively for neurology — exception: children with dysfunctional voiding (Hinman syndrome).**

COMPARISON — BLAIVAS TYPES OF DSD

Type	EMG pattern	Associated lesion	Treatment need
1	Sphincter activity rises with detrusor pressure, relaxes suddenly at peak → unobstructed voiding	Incomplete	Observe unless persistent VUR, hydronephrosis or autonomic hyperreflexia
2	Sporadic sphincter contractions throughout detrusor contraction	Complete	Treat
3	Crescendo-decrescendo contraction → obstruction throughout	Complete	Treat

- Alternative: **intermittent vs continuous** (continuous → complete injury); type matters less than **surveillance and prompt treatment; no link with level.**
- **Pseudodyssynergia** — EMG rise with pressure without true DSD: (1) **abdominal straining** (to initiate/augment or from discomfort); (2) **attempted inhibition** of an involuntary or uncomfortable contraction. Misdiagnosis leads to harmful treatment.
- **Untreated, ~half of men develop VUR, upper tract deterioration, stones, urosepsis, ureterovesical obstruction;** far less in women (lower pressures).
- **DLPP ≥40 cm H₂O after sphincterotomy = failure** (upper tract damage, persistent DSD) — applies to other treatments.
- **Treatment:** oral sphincter drugs disappoint. Options: (1) **CIC + DO control (preferred)**; (2) **sphincterotomy**; (3) **sphincteric onabotulinumtoxinA**; (4) **indwelling catheter**; (5) **diversion**. **Urethral stents abandoned (complications; withdrawn).** **Sphincterotomy (surgical or chemical) only after counselling, for those unable/unwilling to use other options** — poor durability, repeat procedures.

Dysfunctional Voiding

- Synonyms: **non-neurogenic neurogenic bladder, occult (neuropathic) voiding dysfunction, learned voiding dysfunction, Hinman syndrome** — **involuntary striated sphincter obstruction without neurologic disease.**
- **Proof requires pressure-flow-EMG showing emptying during involuntary sphincter contraction with no abdominal straining;** many reported cases rest on weaker evidence.
- Cause uncertain: **persisting transitional phase of continence development, or learned reaction to voiding discomfort** outlasting its cause.
- **Non-neurogenic chronic retention (consensus): PVR ≥300 mL persisting ≥6 months on ≥2 occasions,** stratified high/low risk and symptomatic/asymptomatic. **High risk** (hydronephrosis/hydroureter, chronic kidney disease, recurrent UTI, UI with skin sequelae) → **drainage** (intermittent or indwelling); **suprapubic preferred to urethral long term;** lifelong follow-up.

Bladder Neck Dysfunction

- **Incomplete bladder neck opening during voiding**; synonyms: **smooth (proximal) sphincter dyssynergia**, **primary bladder neck obstruction**, **dysfunctional bladder neck**. "Smooth sphincter dyssynergia" = the neurogenic version with **autonomic hyperreflexia**.
- **Non-neurogenic form: young and middle-aged men with long-standing voiding and storage symptoms**, often labelled **psychogenic** (normal prostate, low PVR, normal cystoscopy); characterised by Turner-Warwick (1973).
- **Differential**: bladder neck contracture, benign prostatic enlargement/obstruction, dysfunctional voiding, prostatitis, neurogenic dysfunction, low-pressure/low-flow.
- **Diagnosis**: **urodynamics proves BOO → localise to bladder neck by videourodynamics, voiding cystourethrography or micturitional profilometry** (or infer from obstruction without stricture, prostate or DSD). DO/↓ compliance coexist in $\frac{1}{3}$ – $\frac{1}{2}$.
- **Cause**: abnormal muscle arrangement (narrowing not funnelling), **sympathetic hyperactivity** (anxious, "high-strung"; partial α -blocker response).
- **"Trapped prostate"** (Turner-Warwick): bladder neck dysfunction + prostatic growth → **double obstruction**; lobes expand into the urethra; lifelong poor voiding accepted as normal.
- **Treatment**: **α -blockers help some; definitive = bladder neck incision**; with trapped prostate → small TURP/ablation including the bladder neck or transurethral incision — patients "have never voided so well".

Bladder Outlet Obstruction in Females

- **Uncommon**; female primary bladder neck obstruction (PBNO) rare.

COMPARISON — DEFINITIONS OF FEMALE BOO

	Nitti (videourodynamic)	Groutz (pressure-flow)	Solomon nomogram
Criterion	Radiographic obstruction bladder neck → distal urethra with sustained detrusor contraction (usually ↓ flow)	Free Qmax ≤12 mL/s repeatedly + Pdet at Qmax ≥20 cm H₂O	Qmax and Pdet@Qmax → probability of obstruction
Fluoroscopy	Required	Not required	Not required
Obstructed	~23% of those studied	~7%	Validated; good agreement with Nitti
PBNO share	~16%	~8%	—
Commonest cause	Dysfunctional voiding , then cystocele, prior anti-incontinence surgery	Prior anti-incontinence surgery , then severe prolapse, stricture	—

- **Surgery for female bladder neck obstruction risks sphincteric UI** — approach cautiously; distinguish dysfunctional voiding from bladder neck dysfunction (uroflow, EMG, fluoroscopy); **multidisciplinary: biofeedback, behavioural, drugs**.
- **High-grade pelvic organ prolapse (POP) obstructs by urethral kinking** (over half); **test voiding with prolapse reduced before repair**; Qmax improves in most after correction.

Low-Pressure and Low-Flow Voiding in Younger Males: Paruresis (Bashful Bladder)

- Low-pressure/low-flow = **decompensating detrusor (usually from BOO) or DO with detrusor underactivity (DO-DU, formerly DHIC/DOIC)**. In young men: frequency, hesitancy, poor stream; endoscopy normal.

- **Paruresis (shy/bashful bladder):** marked hesitancy in the presence of others; 3–19% of young men referred for urodynamics.
- **Obsessional rather than anxious**, over-control of micturition → **behavioural modification**. Ambulatory urodynamics finds a surgically correctable cause only in men >40; negligible yield ≤40. Comorbid **anxiety and depression** common → psychological/multidisciplinary care.

Urinary Retention: Fowler Syndrome in Young Females

- Female retention **unusual but not rare**; causes neurologic, pharmacologic, anatomic, myopathic, functional, psychogenic.
- **Fowler syndrome:** retention in young women (<30; 20–35) without overt neurologic disease — unable to void ≥1 day, **no urgency, rising discomfort**; capacity >1 L without urge required; normal brain/cord MRI; **polycystic ovaries** frequent.
- **Hallmark:** concentric needle EMG of the urethral sphincter — **complex repetitive discharges and decelerating bursts** → impaired relaxation; hypothesised **hormonally dependent channelopathy** (hence premenopausal); **urethral closure pressure very high (>100 cm H₂O)**, enlarged sphincter volume. Urethral afferent overactivity inhibits bladder afferents (**exaggerated cortical pro-continence activity**).
- **Urodynamics: detrusor acontractility.** EMG pattern occurs in ~8% of routine female studies; link to complete retention is rare.
- **Hormonal, drug and onabotulinumtoxinA therapy fail. SNM is the treatment — success ~70%**, even after long retention; **Fowler syndrome is a positive predictor of SNM response** vs idiopathic retention.
- Presumptive cases: confirmed in ~⅓; a third undiagnosed. Psychological comorbidity (somatisation, depression, anxiety, trauma, obsessive-compulsive disorder) overlaps with idiopathic retention → **multidisciplinary care**.

Postoperative Urinary Retention

- Incidence **5–70%**; commonest after **LUT, perineal, gynaecologic and anorectal surgery**.
- **Contributing factors:** 1. traumatic instrumentation; 2. bladder overdistension; 3. ↓ bladder sensation; 4. ↓ contractility; 5. ↑ outlet resistance; 6. ↓ micturition reflex activity; 7. **nociceptive inhibitory reflex**; 8. pre-existing outlet pathology (BPH). **Anaesthesia/analgesia affect 2, 3, 4, 6**; a sympathetic nociceptive reflex could explain 4, 5, 6.
- **Prevention:** **18–24 h catheter decompression after joint replacement lowers retention vs CIC** without more infection; **avoid acute overdistension** (reduces the detrusor response during and after distension).
- **Phenoxybenzamine prophylaxis** reduces retention (meta-analysis); tamsulosin before spinal surgery did **not** (randomised trial).
- **After spinal surgery exclude epidural haematoma → cauda equina syndrome** (saddle anaesthesia, leg weakness).

Hyperthyroidism

- **Sympathetic overactivity/autonomic imbalance** → LUTS, especially in **women**: incomplete emptying, frequency, straining, ↓ flow, enuresis; ↑ PVR, large capacity, ↑ perineal EMG on voiding.
- **Resolve with treatment of thyrotoxicosis**; no correlation with hormone levels.

Schizophrenia

- More **UUI and bed-wetting** than mood-disorder controls; involuntary contractions with childhood incontinence history → **possible brain pathology**; **antipsychotics** add UI via **α-blockade and hypodopaminergic state** (urgency, enuresis, DO).

Gastroparesis

- Delayed gastric emptying without obstruction (diabetic, post-surgical, idiopathic); abnormal contractions and delayed sensation common. **Idiopathic → difficulty emptying; diabetic → frequency**; suggests a **common autonomic neuropathy**.

Myasthenia Gravis

- **Autoantibodies to nicotinic acetylcholine receptors** → striated weakness. **Greatly increased incontinence after prostatectomy** (any disease lowering distal sphincter tone does this).
- De novo dysfunction (intrinsic sphincter deficiency, DO, areflexia) may mark an autonomic subset with worse prognosis. **Intradetrusor toxin and PTNS used without neurologic worsening**.

Isaacs Syndrome

- **Continuous muscle activity** (fasciculation, myokymia, sweating, ↑ creatine kinase), **anti-potassium channel antibodies**; retention from **striated sphincter spasm** (catheter will not pass) + rectal spasm → **plasmapheresis, muscle relaxants, suprapubic drainage**; resolves.

Wernicke Encephalopathy

- **Thiamine (B1) deficiency**, alcoholic or not; periventricular lesions (mammillary bodies, medial thalamus, hypothalamus, cerebellar vermis, PAG, midbrain tegmentum); thiamine deficiency also causes peripheral neuropathy.
- **Urgency UI with DO and small capacity** (medial thalamic–hypothalamic/PAG lesions), or **poor contractility with ED** in alcoholism — **both resolve with thiamine**.

Systemic Sclerosis (Scleroderma)

- Collagen overproduction → skin, vessels, gut, heart, lung, kidney. **Areflexia ± ↓ compliance with detrusor collagen deposition**; other series find few abnormalities and no correlation with fibrosis — **fibrotic replacement ± autonomic dysfunction**.

Ehlers–Danlos Syndrome

- Inherited connective tissue disorders (>10 subtypes): skin fragility/hyperextensibility, joint mobility. **Bladder diverticula (higher recurrence after repair) and VUR**; compliant diverticular tissue (matrix change).
- Female survey: **very high SUI (young women), UUI, pelvic pain, POP, faecal incontinence** — **urethral hypermobility** from collagen defect; UI worst for QoL.

Myotonic Dystrophy

- **Autosomal dominant** myotonia, distal atrophy, cataracts, endocrine, cognitive, testicular, alopecia, conduction defects. **No sphincter myotonia; no characteristic pattern; symptoms do not predict urodynamics** → **study before treating**.

Corticobasal Degeneration

- Rare, **unilateral**, cortex + basal ganglia; urinary symptoms common (1–3 years after onset), **nocturia first**; **DO commonest**, small capacity, hypocontractility, low compliance; **no DSD**.

Sacroccygeal Teratoma

- **Commonest congenital neoplasm**; post-resection urologic complications in ~1/3 (neurogenic bladder, UI, VUR); **prenatal hydronephrosis predicts sequelae**.

Williams–Beuren Syndrome

- **Chromosome 7 deletion** (cardiovascular, facial, intellectual disability). **LUTS ~80%** (urgency, frequency, UII); **bladder diverticula commonest anomaly**; DO common, some DSD, ↓ capacity.

Amyloidosis

- **Hereditary transthyretin amyloidosis (ATTR; formerly type 1 Portuguese familial amyloid polyneuropathy)**, autosomal dominant: early autonomic/genitourinary involvement; ↓ **sensation and DUA (amyloid infiltration of nerves)** even in asymptomatic carriers; **SUI and ↓ anal tone**; LUTS in >80%.

Machado–Joseph Disease

- **Spinocerebellar ataxia type 3 (SCA3)**, commonest dominant hereditary ataxia (MJD1 expansion, 14q32.1): spasticity, rigidity, dystonia, ophthalmoplegia, eyelid retraction. LUT dysfunction ~14%: **urgency, DO, delayed sensation, ↑ PVR**.

Radiation

- **Early reaction (peak 4–6 weeks)**: storage symptoms, ↓ first desire, ↓ capacity, ↓ compliance → **recovers by 6 months**.
- **Late effects**: less common but **progressive, intractable**; ↓ capacity, DO (up to ⅓), ↑ filling pressure. Rat model: **biphasic compliance loss (4–6 and from 10–12 weeks)** — **fibrosis did not explain it**; smooth muscle degeneration, **unmyelinated axon degeneration**, mast cells did.
- Prostate external beam: capacity falls modestly; symptoms correlate with **bladder dose**. Men with SUI after radiation: **more DO, smaller capacity**, more storage treatment. **Combined-modality prostate radiotherapy: long-term reconstruction or diversion in a substantial minority**.
- **Pelvic radiation is an independent LUTS risk factor in women** (frequency, urgency, SUI, ↓ compliance after radical hysterectomy + radiation).

Defunctionalised Bladder

- Early fetal diversion: ↓ weight, connective tissue infiltration, smooth muscle disorganisation, ↓ carbachol response.
- **Previously normal defunctionalised bladder → ↓ capacity, involuntary contractions and/or ↓ compliance**; abnormal bladders keep prior pathology plus these. **Rehabilitate by cycling with increasing volumes**.
- **Renal transplantation into a defunctionalised bladder is feasible without augmentation** — threshold proposed: **capacity ≥100 mL and voiding pressure ≤100 cm H₂O** during rehabilitation; paediatric series and a randomised trial (cycling vs none) show recovery without graft compromise. Still **assess bladder function before transplant**.

Disorders of Sexual Differentiation

- **Congenital adrenal hyperplasia (CAH)** (after feminising surgery) → **more UII and SUI**; another series (CAH, androgen insensitivity) found LUTS common but UI not increased — **effect uncertain**.

Aging

- Autonomic axonal dystrophy; ↓ **detrusor muscarinic receptors**; healthy women show **age-related ↓ contractility, ↓ filling sensation, ↓ MUCP**. Age cannot be separated from coexisting disease.
- **Occult neurology** (multiple cerebral infarcts, cervical and lumbar spondylosis) masquerades as BOO (Olmsted County) → **urodynamics in older men with complex LUTS**.

Benign Joint Hypermobility Syndrome

- Inherited connective tissue disease; **UI and anal incontinence more frequent**, childhood history; **VUR frequent in children**; attributed to collagen metabolism.

Attention-Deficit/Hyperactivity Disorder

- **3–5% of children**; more **enuresis, urgency, holding manoeuvres**, voiding symptoms; continence achieved later; treatment of ADHD may improve LUTS.

Other Conditions

- **Visceral cross-talk**: altered afferent input from other pelvic organs (interstitial cystitis, irritable bowel); experimental colitis causes **reversible detrusor contractility change without histologic change**.
- **Further conditions with reported LUT dysfunction**: adrenoleukodystrophy/adrenomyeloneuropathy; adult polyglucosan body disease; Behçet disease; Brown-Séquard syndrome; central cord syndrome; cystic fibrosis (incontinence); Down syndrome; Duchenne muscular dystrophy; familial dysautonomia; intramedullary epidermoid cyst; Lambert-Eaton myasthenic syndrome; lateral meningocele (Lehman) syndrome; neurofibromatosis; pituitary adenoma; Sjögren syndrome; spinal cord tumours; spinal muscular atrophy.

Concept — *these conditions are grouped because the **bladder behaves as the lesion predicts, not as the diagnosis suggests**: true DSD demands a pons-to-sacral-cord lesion, female and young-male "obstruction" is often functional, and a sphincter already weakened by neuromuscular disease (MSA, myasthenia, radical pelvic surgery) turns outlet surgery into incontinence.*

EVALUATION, SURVEILLANCE, AND TREATMENT OF NEUROGENIC LOWER URINARY TRACT DYSFUNCTION

Overview

- Practice varied widely for lack of evidence; now guided by the **AUA/SUFU Guideline on NLUTD (2021)** and **EAU Guidelines on Neuro-urology (2022)** — still limited evidence.
- **Gaps**: sexual and bowel dysfunction, urologic cancer screening, transitional care, fertility.

Initial Evaluation

- **Risk stratification** (AUA/SUFU) — **prevention of renal damage is the primary goal** (renal failure was the leading SCI death before World War II).
- **Minimum for all: history, physical examination, urinalysis.**
 - History: neurologic condition (onset, level, disability, prognosis), current and past bladder management, symptoms and bother, **goals of care**, prior surgery.
 - Examination: cognition, mobility, scars/stomas/tubes, pelvic examination (women) or digital rectal examination (men), **spasticity, sensation, hand dexterity, contractures**, perineal/buttock skin and pressure sores.
 - Urinalysis interpreted in light of catheterisation. **PVR in anyone voiding spontaneously**; consider diaries, uroflowmetry, patient-reported outcome measures.
- **Risk by lesion**: suprapontine (DO, low PVR) → lower risk; elevated PVR → moderate; sacral/infrapontine → acontractile, low pressure but **compliance may fail**; **infrapontine-suprasacral** → **DO + DSD** → **unclear risk** → **urodynamics**.

TABLE 109.2 RISK STRATIFICATION OF PATIENTS WITH NEUROGENIC LOWER URINARY TRACT DYSFUNCTION (ADAPTED FROM THE AUA/SUFU 2021 GUIDELINES)

Risk	Typical lesion	LUTS	Urodynamics	Imaging	Renal function
Low (must meet all)	Suprapontine, or distal to cord/peripheral, uncomplicated	Spontaneous voiding, low PVR, storage symptoms, stable	(If done) synergic voiding, DO	(If done) normal	Normal/stable
Unknown	Infrapontine–suprasacral; sacral/infrasacral	Storage + voiding; any change; ↑ PVR; recurrent UTI	Not yet assessed	Any abnormality: hydroureteronephrosis, stones	—
Moderate	—	—	Neurogenic retention; DO with incomplete emptying	Normal	Normal/stable
High (any one)	—	—	Poor compliance, ↑ storage pressure, DO, DSD, VUR	Hydroureteronephrosis, scarring, parenchymal loss, large/increasing stones	Abnormal/unstable

- Anyone not clearly low risk is "unknown" until urodynamics, upper tract imaging and renal function reclassify them as moderate or high.
- Low risk needs no routine imaging, renal function or urodynamics unless results would change diagnosis, prognosis or treatment.
- **Trap:** DO alone is compatible with low risk (suprapontine, synergic, low PVR); **DO in the high-risk row means DO with the pressure consequences of infrapontine disease.**
- **Cystourethroscopy not routine** — for haematuria, stricture/false passage, suspected stones or anything altering management.
- **Defer full assessment during spinal shock or acute brain injury (cerebral oedema); no irreversible intervention on symptoms during this unstable phase.**

Management of Neurogenic Lower Urinary Tract Dysfunction: Surgical and Nonsurgical

- Therapies are classified by **function** — **facilitate storage vs facilitate emptying** — and by **target** — **bladder vs outlet**.

BOX 109.2 VOIDING DYSFUNCTION: GOALS OF MANAGEMENT

- **Upper tract preservation/improvement; absence or control of infection; adequate storage at low pressure; adequate emptying at low pressure; control over micturition; no catheter or stoma; social and vocational acceptability.**

BOX 109.3 REASONS TO CHANGE OR AUGMENT A GIVEN REGIMEN

- **Upper tract deterioration; recurrent sepsis/urinary fever; lower tract deterioration; inadequate storage, emptying or control ("inadequate" includes high detrusor pressure, not just volume); unacceptable side effects; skin changes from incontinence or appliances.**

BOX 109.4 PATIENT FACTORS TO CONSIDER WHEN CHOOSING THERAPY

- Prognosis (progressive/malignant disease); general health; **limiting factors (hand dexterity, transfers, habitus)**; mental status; motivation/compliance; wish to be catheter-free or avoid surgery; sexual activity; reliability; educability; family support; economic resources.

- **Principles:** results need not be perfect — **satisfaction and avoidance of adverse outcomes**; decisions shared with patient and carers (reversibility, side effects, best/worst case, follow-up, alternatives); **start with the simplest, most reversible therapy and climb; the patient decides when "enough is enough"**; combine treatments with different mechanisms and non-additive side effects.
- **Emptying:**
 - **CIC is the gold standard and foundation** for the willing patient/carer; **keep catheterised volumes <500 mL**.
 - **Credé/Valsalva not recommended** (outlet resistance).
 - **If chronic catheter needed: suprapubic > urethral** — similar UTI, stone and cancer rates, but urethral erosion avoided; patients prefer it.
 - CIC vs indwelling urethral: **less pyelonephritis, epididymo-orchitis and urosepsis; highest symptomatic UTI with indwelling urethral catheters; lowest with spontaneous voiding**.
 - **Hydrophilic-coated catheters lengthen time to first UTI** (randomised trial; systematic review).
 - **QoL does not always favour CIC** — some prefer indwelling catheters or surgery; dependence on carers harms independence → individualise.
 - **inFlow device** (magnetic intraurethral valve-pump, women): improves QoL but high dropout; for those unable to catheterise.
- **Storage: medical therapy first-line for NDO** (antimuscarinics most studied; **mirabegron** effective); refractory/intolerant → **intradetrusor onabotulinumtoxinA** (~60% continue long term).
- **DO + DSD: detrusor relaxation (drugs or toxin) + CIC preferred**; sphincterotomy for selected men (high failure, repeat procedures); **intrasphincteric toxin not routine**.
- **Neuromodulation:** dorsal penile/clitoral nerve stimulation suppresses NDO; pudendal stimulation (neurophysiologic guidance) ↑ capacity; **sacral deafferentation + anterior root stimulator (Brindley)** — high continence, fewer UTIs, stable kidneys, autonomic dysreflexia resolves; rhizotomy benefits even if stimulator switched off. **PTNS for spontaneous voiders with OAB (MS, PD, stroke); SNM for NLUTD without SCI or spina bifida** (MRI-compatible devices).
- **Future:** lumbar-to-sacral ventral root rerouting, regenerative medicine — unstandardised.
- **Last resort — reconstruction/diversion** (e.g. unable to catheterise): **bladder neck closure + suprapubic catheter** (retropubic better than perineal — fewer fistulae; catheter care determines success); Mainz pouch or augmentation + fascial sling (both acceptable); **ileovesicostomy** (robotic: longer, costlier). **Reserve for failure of less invasive options, after discussion of goals.**

Urinary Tract Infections and Antibiotic Use

- **UTIs near universal in long-term SCI**; bacteriuria common in the first year. **Risk factors: male sex, cervical injury, condom catheter.** Recurrence suggests **calculi, silent pyelonephritis or high residuals**.
- **Treat bacteriuria only when symptomatic** (consensus since 1992).
- **Recommendations:**
 1. Treat bacteriuria only if symptomatic.
 2. Choose agents with little collateral damage to normal flora.
 3. **Treat ≥5 days; 7–14 days for reinfection or relapse.**
 4. Correct structural and functional risk factors.
 5. **Prophylaxis only for recurrent UTI with no treatable cause, especially with dilated upper tracts.**
 6. **No prophylaxis for indwelling catheter or intermittent catheterisation users** (current guidelines agree).

- **Refractory recurrent UTI → upper tract imaging, cystoscopy, urodynamics.**

Neurogenic Bladder and Bladder Cancer

- **Historical link with long-term catheterisation: squamous cell carcinoma** prominent (sometimes invisible endoscopically — found on biopsy), squamous metaplasia, leucoplakia; **risk factors: indwelling catheter, bladder stones, chronic UTI.**
- **Proposed features in SCI:** higher relative risk; higher squamous proportion; **younger age at peak; advanced stage at diagnosis; neither cystoscopy nor cytology reliable;** aggressive evaluation for multiple risk factors.
- **Counterview:** large multinational SCI data and spina bifida data show **incidence comparable to the general population** (meta-analysis ~0.3%), but **many present muscle-invasive → higher mortality; transitional cell carcinoma commonest** in contemporary data; **catheters and UTI, not SCI, are the risk factors.**
- **Screening: AUA/SUFU 2021 does not recommend surveillance cystoscopy** in NLUTD, catheterised or not — insufficient evidence; **overdetection of benign lesions → biopsies and harm. Cystoscopy for haematuria, cancer history or other indications.**

Follow-Up

- Mortality falling → **lifelong urologic follow-up;** long-term SCI cohorts: **every patient has UTIs;** commonest other complications **stones, hydronephrosis, VUR;** also pyelonephritis, dysreflexia, urethral injury, urosepsis.

COMPARISON — FOLLOW-UP RECOMMENDATIONS FOR NLUTD

Body	Frequency	Content
American Paraplegia Society (SCI)	Annually for 5–10 years, then every other year if well	Upper and lower tract evaluation and urodynamics at same intervals; nuclear renogram: >20% fall in renal plasma flow → investigate; annual cystoscopy with indwelling catheter
Consortium for Spinal Cord Medicine (2006)	Yearly	No evidence on optimal tests or interval
EAU	Specialist review every 1–2 years (more often in MS, acute SCI, unstable disease)	Dipstick at least 2-monthly and when UTI suspected; ultrasound (upper tract, bladder) + residual every 6 months; examination and bloods yearly
AUA/SUFU 2021	Risk-based	Low risk: no routine surveillance. Moderate/high: yearly history, examination, symptoms and renal function; upper tract imaging every 1–2 years; repeat urodynamics for change; stone history/risk: stone imaging every 1–2 years; bowel reconstruction: yearly history, examination, electrolytes/acid-base, renal function, upper tract imaging

- **Pattern:** older schedules were **calendar-based and SCI-specific;** AUA/SUFU is **risk-based** and exempts low-risk patients. **Trap:** annual cystoscopy for catheterised patients (APS) is **not** endorsed by AUA/SUFU 2021.

Concept — every evaluation and treatment decision serves one aim, **protecting the kidneys from sustained high bladder pressure**, so risk is judged by pressure, compliance and upper tract status, not by the diagnosis. Treatment then climbs from the most reversible option, and surveillance intensity follows risk.